Board Specs

- OMAP3530 Processor
  - ARM 600MHz ARM Cortex-A8 (>1200 DMIPS)
  - Over 1 billion dhrystones per second with NEON and VFP extensions
  - 2D/3D graphics accelerator
  - C64x+ DSP and video acceleration
- 256MB DDR @166Mhz
- 256MB NAND
- High Speed USB 2.0 OTG
- High Speed USB HOST
- S-Video (TV out)
- DVI-D
- Stereo Audio Out & In
- 8-bit SD/MMC
- JTAG
- UART
- Expansion connector to provide extended connectivity with standard connections like:
  - I2C, MMC, SPI, GPIO
  - LCD expansion—The new LCD interface can be used with an adapter board to connect to displays, including RGB LCD panels and LVDS panels.
  - PWM—The expansion connector now provides three pulse width modulation (PWM) signals for motor driver functions or PWM signalling, typically used in robotics.

Syed quips, "The RAM is fixed for 256 MB and is not expandable; therefore memory requirement is a constraint to run all these applications in parallel. But it's a board limitation rather than one related to the processor. A cell phone-like battery is sufficient to support all these applications, but definitely not for hours when run in parallel. The OMAP3530 chip supports ~ 82hrs of audio (1400mAh) and ~ 4hrs of video (1400mAh)."

Moreover, the Beagle Board is also provided with a S-Video and a DVI-D port. They can be used for connecting to a TV and a monitor respectively. The UART is a serial port and it helps a developer for kernel-level debugging. The JTAG cable that is provided can be used for JTAG-based simulator and low-level debugging. The device also has an 8-bit SD/MMC, which can be used as a hard disk. The PWM expansion connector provides three pulse width modulation (PWM) signals for motor driver functions or PWM signalling, typically used in robotics.

Now, what about the scope for expandability of the board? The board is designed to use standard (PC) peripherals like a DVI, keyboard, mouse, Ethernet, Bluetooth, Wi-Fi, hard disk (over USB), etc. You can upgrade/change/expand these peripherals based on need. The board also provides an expansion header that provides OMAP3530 pin outs to interface with any external device over SPI, MMC, I2C and GPIO interfaces. A few examples are:

1. Beagle is used by an open community member to build an open software defined radio—www.opensdr.com
2. The expansion header can be used to connect this to an ADC (analogue to digital converters), and process the live samples on DSP and use ARM for the host interface.
3. TI DLP Pico projector can be interfaced through the DVI interface on Beagle Board and the framebuffer content can be projected [*dkc1.digikey.com/us/mkt/pico.html*]
4. With Angstrom distribution ported, the board can be used as a regular desktop.
5. Android is supported by BeagleBoard.org community.
6. u-buntu (the Canonical port for ARM Cortex A8) will bring in thousands of apps on Beagle Board.

The bottom line is that, as a developer, you can use it for Web services, 3D gaming, for the Linux kernel and drivers, boot loaders and firmware, portable media and infotainment applications, the UI framework, high-end signal processing with generic DSP, OLPC applications like Sugar— in short, the possibilities are endless; you'll only have to let your imagination run wild. Just to give you an idea, you might want to check out all the current projects that are under way at *www.beagleboard.com/projects*. So, what are you waiting for? Take a bite at this 'Open Platform'.

Syed notes, "While there are numerous theoretical concepts available today, these could be executed with 'Open Platforms', which is a new trend in the IT industry where products are defined, designed and developed by consumers across the world. Ideas are just like raw materials. They need enormous energy, effort and tools to convert them into end products. Collaboration in such efforts results in better execution." END

References and documentation

- Beagle Board community portal: *beagleboard.org*
- Technical reference manual for OMAP3530 processor: *focus.ti.com/general/docs/gencontent.tsp?contentId=36915*
- DSP LINK (for ARM-DSP communication): *www-a.ti.com/downloads/sds_support/targetcontent/link/link_1_50/index.html*
- Validation software, Linux device drivers, latest kernel from GIT: *code.google.com/p/beagleboard/wiki/BeagleboardRevCValidation*
- Board schematics: *beagleboard.org/static/BBSRM_latest.pdf*
- DSP Compiler: *www-a.ti.com/downloads/sds_support/targetcontent/LinuxDspTools/download.html*
- ARM Compiler: *www.codesourcery.com/gnu_toolchains/arm/portal/release313*
- More links and documentation: *elinux.org/BeagleBoard & code.google.com/p/beagleboard*
- Discussion list: *discussion at beagleboard dot org*
- IRC: *#beagle on irc.freenode.net*

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While Atanu likes to head bang and play air guitar in his spare time, Abhijit loves to hack on open source and is a gamer by heart. Oh, and they're also part of the LFY Bureau.